ANALYSIS OF ATTENTION MECHANISMS FOR HORIZON LINE DETECTION AND CAMERA ORIENTATION ESTIMATION

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ABSTRACT

The geometric characterization of the external parameters of cameras requires labor-intensive methods during their construction and field execution, where measuring multiple points in images is essential to ensure redundancy and reliable results. However, conventional characterization is cumbersome and demands specific data collection. Recently, efforts have focused on deep learning-based solutions to replace repetitive manual tasks. Various learning strategies, neural networks, geometric priors, and datasets have been investigated. This paper analyzes the performance of a camera characterization technique based on artificial intelligence and attention mechanisms, highlighting its advantages, limitations, and potential applications to enhance image-based navigation results.

Key words – *Geometric Characterization, Deep Learning, Attention Mechanisms, Transformer.*

1. INTRODUCTION

The characterization of a camera plays a fundamental role in obtaining more precise geometric information, especially in applications involving computer vision, aerial photogrammetry, remote sensing, and robotics—fields with a long history of research and application [1–5]. One of the initial steps in this process is often the characterization of the camera's extrinsic parameters (rotation and translation) [6–8].

However, geometric camera characterization remains a crucial and challenging step for many applications, such as those mentioned above, as it aims to infer intrinsic and extrinsic camera parameters through the analysis of geometric distortions in image sets, employing a specific mathematical model [9].

The characterization of a single-image camera consists of inferring its intrinsic and extrinsic parameters from the analysis of the distortion caused by perspective projection in an input image. This is a central problem in various computer vision applications, such as image rotation correction [7, 8], photo vertical alignment [10, 11], metrology [12, 13], visual aesthetic assessment [8, 14], and object composition for augmented reality [15–17].

The geometry underlying monocular vision is inherently linked to projective geometric cues such as vanishing points (VPs) and the horizon line, which correspond to the intersections of parallel lines and planes in the real world, respectively [18, 19]. A classical approach for single-image camera characterization involves the initial detection of line segments in the input image, followed by the identification of segments representing VPs and the horizon line, using algorithms such as RANSAC or other sampling strategies [10, 20–22]. However, this approach can fail when there is incorrect detection of internal lines, compromising the accuracy of the process.

Several deep learning-based approaches have been recently proposed to overcome these limitations [23, 24]. These methods directly infer camera parameters from a single input image, leveraging semantic cues learned by deep neural networks. While offering more accurate results compared to classical calibration methods, these networks need to learn image geometric structures without explicit supervision, which limits their performance. Recently, some deep learning approaches that combine geometry and semantic cues have been suggested [25, 26], achieving superior results. However, strategies for exploiting lines are still limited to post-processing [25] or the construction of multiple separately trained networks [26].

Furthermore, current neural network-based approaches rely on convolutional neural networks (CNNs), which exhibit limitations in capturing long-range dependencies and, consequently, in extracting global image features such as camera parameters.

To address these limitations, the use of transformers [27, 28] has been proposed, recently adopted in computer vision tasks [29, 30]. Transformers have shown suitability for two primary objectives in single-image camera characterization: (i) efficiently integrating geometric and semantic cues, and (ii) effectively learning relationships and global contextual information present in the image.

This work proposes a Transformer for camera geometric characterization with line classification, utilizing an end-to-end neural network to determine the horizon line for camera orientation, directly estimating camera parameters from an image and a set of line segments [31].

The methodology aims to demonstrate superior performance compared to previous-generation convolutional network methods.

2. MATERIALS AND METHODSS

2.1. Target camera modeling

Numerous studies employ mathematical models to represent cameras, describing the image capture process from a point in 3D world coordinates to its projection onto a 2D image plane. Various camera types and systems correspond to distinct parametric models. This section initially presents a detailed formulation of the basic pinhole camera model. Figure 1 (b)

illustrates the research subject, the camera model targeted in this study.

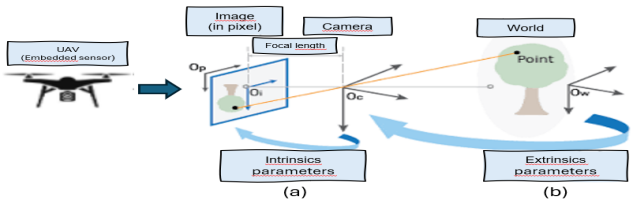


Figure 1: Camera model and systems involved in geometric calibration.

In this model, the coordinate systems involved in the process (the camera coordinate system and the object-space system), as well as the intrinsic characterization parameters, Figure 1(a), and the extrinsic parameters, Figure 1(b), which are the focus of this work, are considered during the geometric characterization process.

2.2. Experimental dataset

The experiments evaluated the performance of the proposed Transformer-based method alongside other benchmark approaches, utilizing datasets derived from UAV (Unmanned Aerial Vehicle) platforms at the Institute for Advanced Studies (IEAv), as well as open-source image databases such as Zioulis et al. [32] and proprietary image banks, as shown in Figure 2.



Figure 2: Example of dataset used in the experiments.

2.3. The architecture of a neural network

Recent neural network-based approaches propose directly predicting camera parameters using semantic cues from an input image learned through convolutional networks. Workman et al. [31] propose the first network-based solution for estimating the horizon line in an image.

In contrast to the studies presented in [25, 26], the present work introduces and analyzes an application of the Transformer that enables the network to perform end-to-end learning, effectively capturing both semantic and geometric information. It has been demonstrated that the proposed auxiliary task of line classification significantly contributes to improving accuracy in camera characterization.

Figure 3 illustrates the utilized architecture, in which, given an input image, the vertical vanishing point (also known as the zenith VP), the horizon line, and the field of view (FoV) related to the camera's focal length are estimated. The employed network consists of the following components: a CNN backbone network responsible for extracting image features; an encoding transformer that processes these features; and a decoding Transformer structure, followed by feedforward networks (FFNs), which generates the predictions for all outputs.

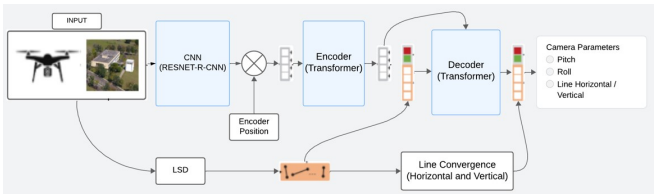


Figure 3: Overview of the neural network architecture applied in the experiments.

The application estimates the parameters necessary for camera characterization from an input image and a set of line segments. Image features are extracted using the ResNet architecture, processed, and positionally encoded before being sent to a Transformer-based encoding network. The learned query vectors are then utilized in a Transformer decoding network, along with line embedding vectors, allowing the queries to capture features from both the image and the line segments. Additionally, two auxiliary outputs—vertical and horizontal convergence line scores—are employed to guide the network in learning the scene's geometry, thereby facilitating the characterization process.

3. RESULTS

Table 1 presents the accuracy of the vertical and horizontal lines estimated by the evaluated Transformer model. Table 2 provides the results of the quantitative evaluation against the presented benchmark.

Table 1: Accuracy of the classification of convergence lines.

Test	Accuracy (%) Vertical	Accuracy (%) Horizontal
ZENODO	99.80	92.05
PRÓPRIAS	90.76	87.15



Figure 4: Examples of the results for horizon line determination.

4. DISCUSSION

The present study analyzed the performance of the Transformer in the geometric characterization of cameras, comparing it with reference methods, including Upright [10], Perceptual [23], and GPNet [26]. The evaluation was conducted using two datasets: ZENODO and a proprietary image database.

Table 2: Comparison of Methods for Geometric Characterization

Method	Pitch Mean (°)	Pitch Med. (°)	Roll Mean (°)	Roll Med. (°)	FoV Mean (°)	FoV Med. (°)	AUC (%)
Upright	2.88	1.70	5.99	0.63	8.90	5.02	75.43
Perceptual	3.09	2.08	1.06	1.00	5.01	4.09	79.10
GPNet	2.02	2.98	0.85	0.57	5.91	4.02	84.02
TRANSFORMER	1.78	1.51	0.60	0.68	4.09	3.32	86.66

In Tables 1 and 2, it can be observed that the Transformer demonstrates superior performance in various aspects. Table 1 highlights the accuracies for vertical and horizontal convergence lines. For the ZENODO dataset, the vertical accuracy reached 99.80%, while the horizontal accuracy attained 92.05%. In the custom dataset, the Transformer achieved a vertical accuracy of 90.76% and a horizontal accuracy of 87.15%. These results indicate significant robustness in the detection of convergence lines, with high generalization even when utilizing data from distinct sources.

Table 2 presents a more detailed quantitative analysis, demonstrating that the Transformer reduces the mean errors compared to the listed methods, such as GPNet, across the following parameters: upward direction, tilt, rotation, and field of view (FoV) angles. Specifically, the reductions were as follows: Pitch Mean: 11.88%, Roll Mean: 29.41%, and FoV Mean: 40.3%, respectively, when compared to the previous State of the Art (SotA), exemplified by the GPNet model. Furthermore, the area under the curve (AUC) of the horizon line errors increased variably from 82.02% to 88.09%, representing an improvement of 7.40%.

Table 2 presents a detailed quantitative analysis, demonstrating that the Transformer significantly reduces the mean errors compared to the Perceptual and Upright methods. When compared to the Upright model, the reductions were 38.19% for Pitch Mean, 89.98% for Roll Mean, and 54.04% for FoV Mean. In comparison to the Perceptual model, the observed reductions were 42.39% for Pitch Mean, 43.40% for Roll Mean, and 18.36% for FoV Mean. These results reinforce the superiority of the Transformer over the reference models in capturing the geometric parameters of the scene.

In comparison to the Perceptual and Upright methods, the AUC of the horizon line errors showed a significant increase, from 79.10% to 86.66%, resulting in an improvement of approximately 9.55% over the Perceptual model. In contrast, when compared to the Upright model, the increase was from 75.43% to 86.66%, representing an improvement of about 14.89%. These trends are consistent across both benchmarks and qualitative results, highlighting the effectiveness of the Transformer in capturing the geometric and semantic features of the scene.

Although the task of classifying convergence lines has been regarded as auxiliary within the Transformer model, the results demonstrate that it plays a significant role in enhancing geometric calibration. The classification accuracy exceeded 90% across both analyzed datasets, reinforcing the model's ability to generalize its learning across different conditions and data sources.

5. CONCLUSIONS

The results indicate that the Transformer offers significant advantages over conventional methods for the geometric characterization of cameras, particularly in extrinsic parameter estimation and convergence line detection. Its attention-based architecture enables the efficient integration of geometric and semantic cues, resulting in notable improvements in the evaluated metrics, such as reduced mean errors and increased AUC.

The utilization of Transformers for the geometric characterization of cameras is promising, demonstrating potential to replace techniques based on Convolutional Neural Networks (CNNs) and other conventional approaches. With improvements in accuracy and generalization capacity, Transformers can be applied in various fields, such as image-based navigation and automatic camera characterization.

Therefore, the proposed methodology not only surpasses previous methods but also paves the way for new approaches in the development of deep learning-based solutions for the geometric characterization of sensors, opening opportunities for future research and advancements.

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